FoveaWAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation

Anonymous CVPR submission

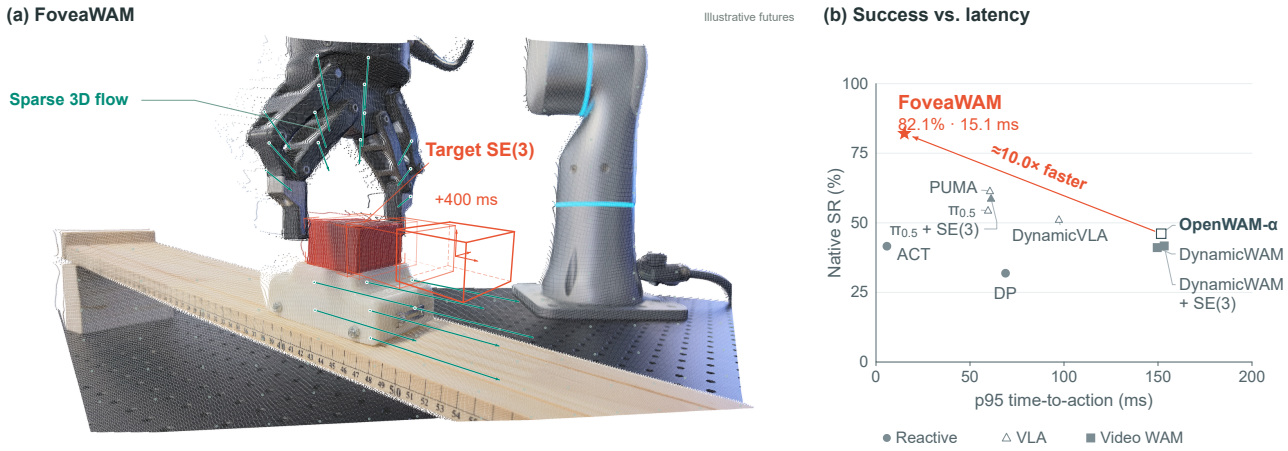


Figure 1. **FoveaWAM allocates prediction precision by task role.** (a) A real-robot scene reconstructed from RGB with MoGe-2 [38], with illustrative target SE(3) futures and sparse 3D flow. (b) Native SR versus p95 time-to-action from Table 1.

Abstract

Manipulating moving objects requires actions that fit the scene when they execute, not when the image was captured, yet every millisecond spent inferring the scene's motion lets the target move further. World–action models forecast future images or latents at uniform fidelity, spending time on appearance while leaving the target's metric pose implicit; target-centric predictors are fast but ignore the surroundings that constrain or drive the motion. We observe that the parts of a dynamic scene need different precision: the target's pose decides contact to within millimeters, the bodies around it matter only through the clearance they leave and the motion they impart to the target, and the remaining background does not matter. We present FoveaWAM, a world–action model built on this role-aware 4D structure. It encodes the target's motion as a metric SE(3) pose history, so one RGB-D frame suffices as input. It learns to forecast the target's future SE(3) trajectory precisely and the whole scene coarsely as sparse 3D flow, with forecasts used only in training. FoveaWAM acts in 15.1 ms on one RTX 5090, about 10× faster than video world–action baselines. We also introduce OnTimeBench, an open-source benchmark of 14 simulated dynamic manipulation tasks in which the scene keeps moving throughout each policy's inference. FoveaWAM achieves 82.1% success on OnTimeBench, 20.7 points above the strongest baseline, leads on 24 DOMINO tasks with 32.5%, and succeeds in 37.5% of trials on six real-robot tasks with diverse target motion.

1. Introduction

Learning-based manipulation has advanced rapidly, from diffusion policies to vision–language–action (VLA) and world–action models (WAMs) [5, 6, 11, 19, 39, 44, 58], yet most of this progress assumes that objects stay still unless the robot moves them. In factories, kitchens, and human–robot handovers, objects keep moving while the robot perceives, computes, and acts. In such scenes, an action must fit the scene at the moment it executes, not the moment its observation was captured, so a policy has to know how the scene moves. That knowledge must itself be inferred, and inference takes time during which the scene keeps moving: the longer a policy computes to anticipate the motion, the further ahead it must anticipate.

Recent dynamic policies add a memory of past motion, typically image-space cues such as optical flow [13, 25], and a forecast of the future [25, 34, 51]. Video-based WAMs forecast whole future frames or latents at uniform fidelity, spending much of their computation on visual detail [39, 44]

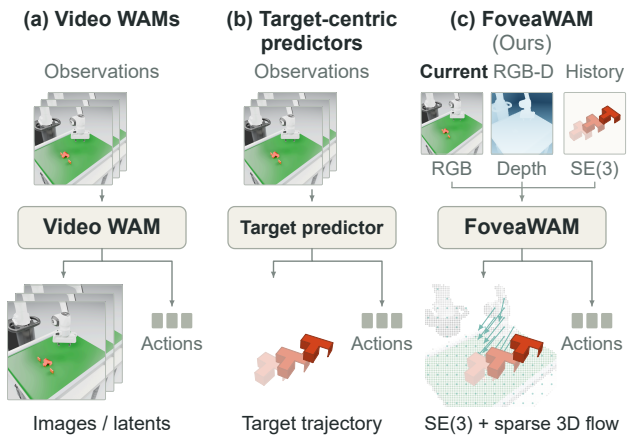


Figure 2. **What dynamic policies forecast.** (a) Video WAMs forecast future frames or latents at uniform fidelity [39, 44]. (b) Target-centric predictors forecast only the target's trajectory [13, 16]. (c) FoveaWAM reads one RGB-D frame with the target's SE(3) history and forecasts the target's SE(3) trajectory precisely and the whole scene coarsely as sparse 3D flow; both forecasts supervise training and are skipped at inference.

(Fig. 2a). Compact future codes [26] reduce this cost and metric 3D scene dynamics [49] add geometry, but both still treat the whole scene uniformly. In each case, the quantity that decides contact, the target's metric pose at execution time, stays implicit. Target-centric predictors and dynamic VLAs with object-level forecasts sit at the other end [13, 16] (Fig. 2b): they are fast but anticipate only the target, leaving out the surroundings that can block or redirect its motion. Neither approach allocates precision according to what contact requires from each part of the scene. This raises a central question: what must a policy know about a moving scene, and how precisely, to make contact while keeping inference latency low?

Our key observation is that a dynamic policy needs to know very little precisely (Fig. 1). Contact demands millimeter precision in the target's pose at execution time; the surroundings matter only through the clearance they leave and the motion they impart to the target, so they need only be known coarsely. Building on this observation, we introduce FoveaWAM, a world–action model that learns a role-aware metric 4D representation, precise where contact demands it (Fig. 2c). To supply motion information without encoding past image frames, FoveaWAM pairs a single RGB-D frame with a short history of the target's metric SE(3) poses. A world stream learns the representation by forecasting the target's future SE(3) trajectory and the motion of the whole scene as a sparse 3D flow. An action stream reads the world stream's scene and summary tokens to generate action chunks, with explicit forecasting confined to training [48]. FoveaWAM thus combines precise anticipation of target motion with low inference latency, limiting how much the scene changes between observation and action.

We build OnTimeBench, an open-source benchmark of 14 simulated dynamic manipulation tasks in which the scene keeps moving throughout each policy's inference, so every method pays for its own latency. FoveaWAM achieves 82.1% success on OnTimeBench, 20.7 points above the strongest baseline, at a p95 latency of 15.1 ms with compiled BF16 inference, about 10× lower than that of the video world–action baselines [25, 39]. It reaches 32.5% on 24 DOMINO tasks [13], 5.6 points above DynamicWAM, and succeeds in 37.5% of trials on six real-robot tasks, against [XX]% for the strongest baseline. Controlled ablations test the contributions of target-motion history and future supervision, and compare role-aware with uniform-fidelity forecasts to examine the allocation of precision.

Our contributions are threefold:

- We show that a dynamic policy needs millimeter precision only in the target's pose at execution time, and instantiate this principle as role-aware 4D modeling: a precise metric SE(3) trajectory for the target and a coarse sparse 3D flow for the whole scene.
- We develop FoveaWAM, a world–action model that combines metric target-pose histories with role-aware predictive supervision for low-latency action generation.
- We introduce OnTimeBench, an open-source dynamic manipulation benchmark that charges each policy its own inference latency, and demonstrate gains over strong baselines on OnTimeBench, 24 DOMINO tasks, and six real-robot tasks.

2. Related Work

Dynamic manipulation. Compensating for the time between sensing and acting is a classical problem, from control with dead time [31] to catching systems that track the target with Kalman-type filters and plan an interception [1–3, 28]; such systems are precise but engineered for each object and task. Learned policies have recently been brought to moving scenes [42]. PUMA reads historical optical flow and is supervised with future object features [13], and DynamicWAM conditions a WAM on flow and kinematic descriptors and executes it with real-time chunking [25]. Both infer past motion from image-space cues, whereas FoveaWAM takes the target's metric SE(3) pose history as input and observes only the current RGB-D frame. Other dynamic policies forecast salient tokens [34], future images [52], compact future codes [26], or 3D scene dynamics [49] and condition their actions on these forecasts at test time. DynaWM adapts a base VLA's action chunks for dynamic manipulation [10], and DynamicManip synthesizes dynamic demonstrations from a static one [21]. Execution schemes reduce or hide latency by overlapping inference with execution [30, 42], keeping consecutive chunks consistent [7, 23], or condition-

ing on the execution time [35]. They decide when actions are computed and applied, not what the policy represents. DOMINO [13] and PhysMani-Bench [49] provide diverse dynamic-manipulation tasks, but their released default evaluation pipelines advance the simulation only after a policy has returned its prediction. OnTimeBench advances the scene during inference and charges each method its measured latency.

Predictive supervision in WAMs. World–action models (WAMs) generate future video and actions in one network [20, 39, 44, 56], building on video prediction as a policy objective [36, 41]. Because generating video is costly, recent work changes what is forecast, such as dynamic regions, depth, and semantic features [51], flow and geometry alongside RGB [47], object masks or slots [24, 45], or compact latents [33, 55], and asks whether the forecast is needed at test time. Fast-WAM retains most of the benefit of video modeling without test-time imagination [48], and PUMA and SG-WAM drop their predictors after training [13, 53]. FoveaWAM keeps this training-only use but forecasts primarily a metric 4D state rather than pixels, features, or latents, so the target's future pose is supervised explicitly instead of being left implicit.

Object-centric and geometric motion representations. Metric 3D inputs make policies more precise and sample-efficient [29, 50]. Motion has been represented by 2D point tracks [4, 40, 43], 3D flow [14, 17, 46, 49], and object keypoints or SE(3) trajectories [16, 27, 32], while rigid-body decompositions have long been used to learn scene dynamics [8, 9]. These works describe the scene with one kind of representation, such as dense flow or tracks everywhere, or a pose for each object. Target-centric predictors, including dynamic VLAs with object-level forecasts, model only the target's future [13, 16] and leave out the bodies that block or push it. DOCIR separates the robot, the target, and obstacles, but uses these distinctions only as input encodings [12]. FoveaWAM assigns precision by role: a coarse sparse 3D flow describes the whole scene, while the target, on which contact depends to within millimeters, is additionally described by its SE(3) history and forecast.

3. FoveaWAM

We formulate dynamic manipulation as generating time-indexed command chunks (Sec. 3.1) from a representation supervised, at the execution times of these commands, with the scene's sparse 3D flow and the target's SE(3) trajectory (Fig. 3). We describe the metric inputs (Sec. 3.2), this supervision (Sec. 3.3), and how an action stream reads the representation to generate the chunk (Sec. 3.4).

3.1. Problem Formulation

We study manipulation in scenes that keep moving while the policy computes. At query t, a policy π learned from demonstrations reads an RGB image I_t and a depth map D_t captured at time τ_t, together with histories $\mathcal{T}_t$ of the target pose $T(\tau) = (R(\tau), \mathbf{p}(\tau)) \in \mathrm{SE}(3)$ and $\mathcal{R}_t$ of the robot state, all in the robot base frame, and outputs a chunk of H commands

$$(\mathbf{a}_{t,0}, \dots, \mathbf{a}_{t,H-1}) \sim \pi\big(\cdot \mid I_t, D_t, \mathcal{T}_t, \mathcal{R}_t\big), \qquad (1)$$

where command j is meant for time $\tau_t + j\Delta_a$, with control period Δ_a.

Each task specifies a *target*, whose pose the robot must match within a contact or containment tolerance. The surroundings that matter form the *context*: obstacles, which the robot must clear, and drivers, which move the target. The chunk becomes available only after inference, so each executed command must account for the target and the context at its execution time rather than at τ_t.

3.2. Metric Scene and Motion Inputs

FoveaWAM encodes the target's motion precisely and its context through the current geometry: the current RGB-D frame gives the metric layout of the scene, and a short history of target poses gives the motion that one frame does not show, which also reflects how drivers have moved the target.

Current scene. FoveaWAM anchors each patch token at a metric 3D point, so that the observation shares the base frame of the forecasts and the actions. A frozen MAE ViT-B/16 [15] turns the RGB image into 196 patch tokens on a 14×14 grid, and the depth map back-projects the center of patch i to a point $\mathbf{x}_i \in \mathbb{R}^3$ in the base frame. A linear embedding of $\mathbf{x}_i$ is added to the patch token, and $\mathbf{x}_i$ drives 3D rotary position embeddings of the patch queries and keys, so that attention between patches depends on their relative 3D positions. The points $\mathbf{x}_i$ are also the source points of the scene flow (Sec. 3.3).

Target motion. Instead of past frames, from which motion must be recovered by comparing images, FoveaWAM receives the target's recent motion as a metric SE(3) state. The history spans the last 350 ms in 8 samples at 20 Hz; its 8 tokens hold the current target pose and the seven motions between consecutive samples, each with its translation, relative rotation, time interval, and finite-difference velocity and acceleration in physical units. The velocity and acceleration that decide where the target will be thus enter the model directly, at the cost of a few low-dimensional tokens; Sec. 4.3 removes this history to measure its effect. In simulation, we use ground-truth target poses from the simulator. On the robot, a causal tracker estimates them from the RGB-D stream.

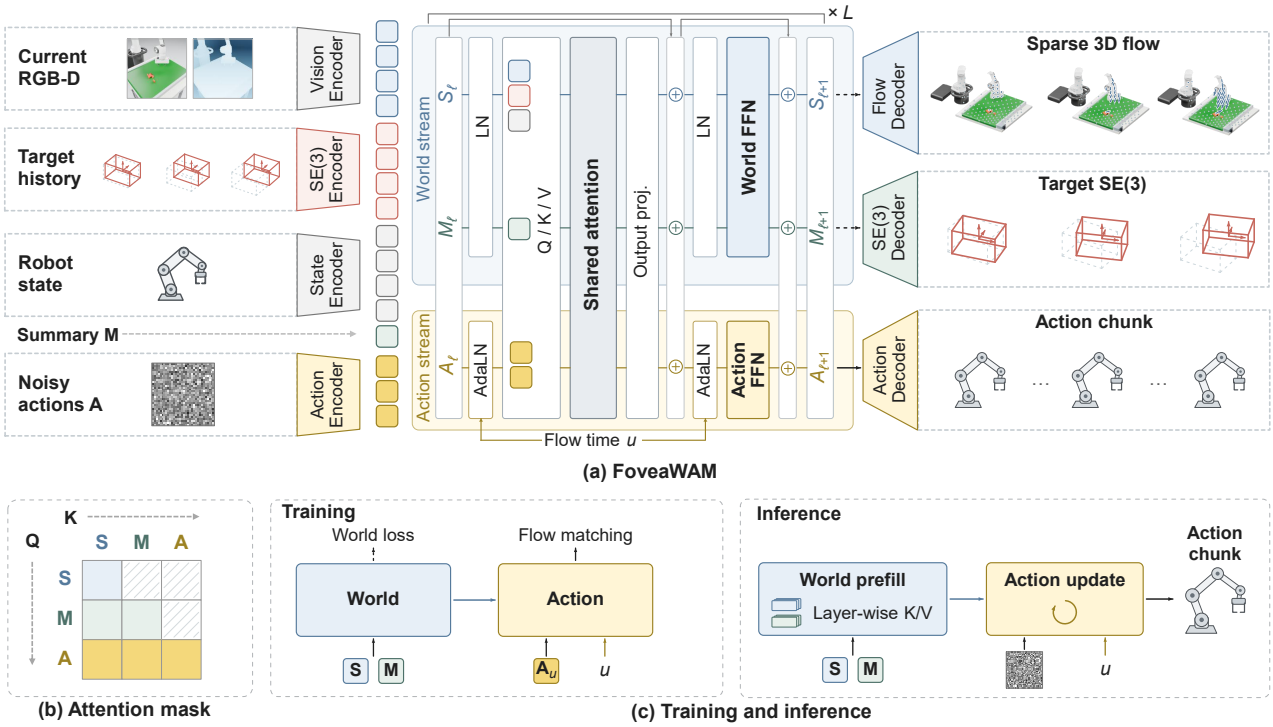


Figure 3. **FoveaWAM architecture.** (a) Current RGB-D and target/robot histories form scene tokens S, alongside learned summaries M and noisy actions A. Each layer shares masked attention, applies one World FFN to S, M and a separate Action FFN to A, and retains both residual updates. Final patch and mean-summary states support training-only geometric readouts (dashed); actions read hidden states. (b) Mask rows are queries and columns are keys. (c) Joint training combines auxiliary supervision and action flow matching. At deployment, world prefill caches layer-wise keys and values for action sampling, skipping geometric readouts. The RGB thumbnail is a Blender rendering. Geometric examples show recorded supervision (flow enlarged); token counts, layer stacking and action icons are schematic.

World stream. The patch tokens, the target tokens, and 8 robot tokens with the end-effector states at the times of the target history form the 212 scene tokens. Together with 16 learnable summary tokens that aggregate them, they make up the world stream.

3.3. Role-Aware 4D Predictive Supervision

FoveaWAM supervises its representation with two geometric forecasts of different precision: the motion of the whole scene as a sparse 3D flow on the patch grid, and, because the target's pose decides contact, the full SE(3) trajectory of the target. Both are expressed in the base frame as changes from the current state at the times $\tau_t + \delta_h$, with $\delta_h = h{\cdot}50\,\text{ms}$ for $h = 1, \ldots, 8$, from 50 ms to 400 ms after capture; together with the current state, these horizons cover the execution times of all commands in the chunk.

Scene flow. At the resolution of the patch grid, the flow covers every surface that the patches see: the robot, the target, and the context, whose clearance and motion the commands must account for. For each of the 196 points $\mathbf{x}_i$ and each horizon h, a linear head on the final state of patch token i predicts the point's base-frame displacement $\hat{F}_{ih} \in \mathbb{R}^3$ and a visibility logit $\hat{o}_{ih}$, trained with

$$\mathcal{L}_F = \frac{1}{|\mathcal{V}_F|} \sum_{(i,h)\in\mathcal{V}_F} \Big[\tfrac{1}{3} \big\| (\hat{F}_{ih} - F_{ih})/s \big\|^2 + \mathrm{BCE}(\hat{o}_{ih}, o_{ih}) \Big], \tag{2}$$

where F_{ih} is the displacement of the surface point at $\mathbf{x}_i$, o_{ih} its visibility, s a fixed scale fitted on the training labels, and $\mathcal{V}_F$ the set of valid labels.

Target trajectory. The flow samples the target only at the few patches it covers, too sparse to determine its rotation, so a dedicated linear head on the mean of the final summary states forecasts its rigid motion directly: for each horizon h, the target's displacement $\hat{\mathbf{d}}_h$ and a rotation increment $\hat{\boldsymbol{\phi}}_h$,

trained with

$$\mathcal{L}_P = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \Big[\|\hat{\mathbf{d}}_h - \mathbf{d}_h\|^2 + \lambda_R \, d_{\mathrm{SO}(3)}\big(\hat{R}_h, R_h\big)^2 \Big], \quad (3)$$

where $\mathbf{d}_h = \mathbf{p}(\tau_t+\delta_h) - \mathbf{p}(\tau_t)$ and $R_h = R(\tau_t+\delta_h)$ are the true displacement and orientation, $\hat{R}_h = \mathrm{Exp}(\hat{\boldsymbol{\phi}}_h^{\wedge}) R(\tau_t)$, $d_{\mathrm{SO}(3)}$ is the geodesic distance, and $\mathcal{V}$ is the set of horizons with valid labels. The rotation term matters for contact: a rotation error θ can displace a contact point at distance r from the target's origin by up to $2r\sin(\theta/2)$. For example, a 5° error at $r = 10\,\mathrm{cm}$ can shift the contact point by nearly 9 mm.

3.4. World–Action Learning and Inference

Action stream. The action stream generates the chunk by conditional flow matching [22]. It holds $H = 8$ tokens, each embedding one noisy command: an end-effector translation increment, a rotation increment, and a gripper command. In every layer, one attention spans both streams with shared projections, followed by separate feed-forward networks; world tokens use layer normalization, and action tokens use adaptive layer normalization conditioned on the flow time. With S, M, and A denoting the scene, summary, and action tokens, the attention mask allows $S \to S$, $M \to \{S, M\}$, and $A \to \{S, M, A\}$, where $X \to \mathcal{Y}$ means that tokens in X attend to tokens in $\mathcal{Y}$. The world stream never reads the actions. The action stream reads, in every layer, both the patch states shaped by scene-flow supervision and the summary states shaped by target-trajectory supervision.

Training objective. For a chunk $\mathbf{a}_1$ normalized with training statistics, noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and flow time $u \sim \mathcal{U}[0, 1]$, the action stream v_ψ regresses the flow-matching velocity

$$\mathcal{L}_A = \mathbb{E}\,\mathrm{MSE}\big(v_\psi(\mathbf{a}_u, u \mid \mathcal{S}_t),\, \mathbf{a}_1 - \mathbf{a}_0\big), \quad (4)$$

where $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$, $\mathcal{S}_t$ denotes the inputs in Eq. (1), and the MSE averages over valid commands and action dimensions. The world and action streams are trained jointly with

$$\mathcal{L} = \mathcal{L}_A + \lambda_F \mathcal{L}_F + \lambda_P \mathcal{L}_P + \lambda_{\mathrm{aux}} \mathcal{L}_{\mathrm{aux}}, \quad (5)$$

where $\mathcal{L}_F$ and $\mathcal{L}_P$ are the flow and pose losses of Eqs. (2) and (3), and $\mathcal{L}_{\mathrm{aux}}$ sums auxiliary appearance losses.

Inference. At deployment, FoveaWAM skips the forecast heads, as recent world–action models do [20, 48, 51]. Because world tokens attend neither to the actions nor to the flow time, the world stream runs once per query, and the action stream integrates 5 Euler steps on the cached keys and values of the scene and summary tokens. Commands are indexed by time from capture, so the policy needs no latency input: the executor hands over each chunk at the first control-period boundary at or after inference completes and starts from the command indexed to that boundary. On the robot, the latency of a query also includes tracking the target in the newest frame. Further implementation details are in the supplementary material.

4. Experiments

We test whether role-aware precision and low latency together improve manipulation when the scene keeps moving during inference. We first compare FoveaWAM with existing policies, each charged with its own latency (Sec. 4.2), then analyze which parts of its design account for the results (Sec. 4.3), and finally verify it on a real robot with tracked target poses (Sec. 4.4).

4.1. Experimental Setup

OnTimeBench. Unlike the evaluators of DOMINO [13] and PhysMani-Bench [49], which advance the simulation only after a policy has returned its prediction, OnTimeBench shares one physical clock among observation, scene motion, and action execution, so the scene keeps moving while a policy computes. It contains fourteen tasks in five families, built in MuJoCo/robosuite [37, 57]: grasping a moving target (harmonic, conveyor, translating, rotating, and occluded motion), placing into a moving receptacle, passing a wand along a rotating wire, capturing rolling or sliding objects, and grasping after an impact (Fig. 4). Experts with privileged simulator state provide 800 demonstrations per task. We select configurations on development scenes and report all OnTimeBench results on 1,400 held-out scenes (100 per task).

Evaluation. During each query, physics advances by the method's inference latency, drawn from its latency distribution measured in its native deployment setting at batch size 1 on one RTX 5090, the GPU of our robot. All methods share one executor, which executes each chunk from the command meant for the control-period boundary at which the chunk arrives. Besides this native setting, we add 150 ms of latency to every method and, as a reference without the cost of computation, pause physics during inference. We report the task-balanced success rate (SR) and the p95 time-to-action, the measured time from an observation to its first command; paired 95% intervals come from a bootstrap over held-out scenes. The supplementary material gives the full evaluation protocol.

Baselines. We compare with the reactive policies ACT [54] and Diffusion Policy [11] and the VLA $\pi_{0.5}$ [5]; the dynamic VLAs DynamicVLA [42] and PUMA [13]; the WAM

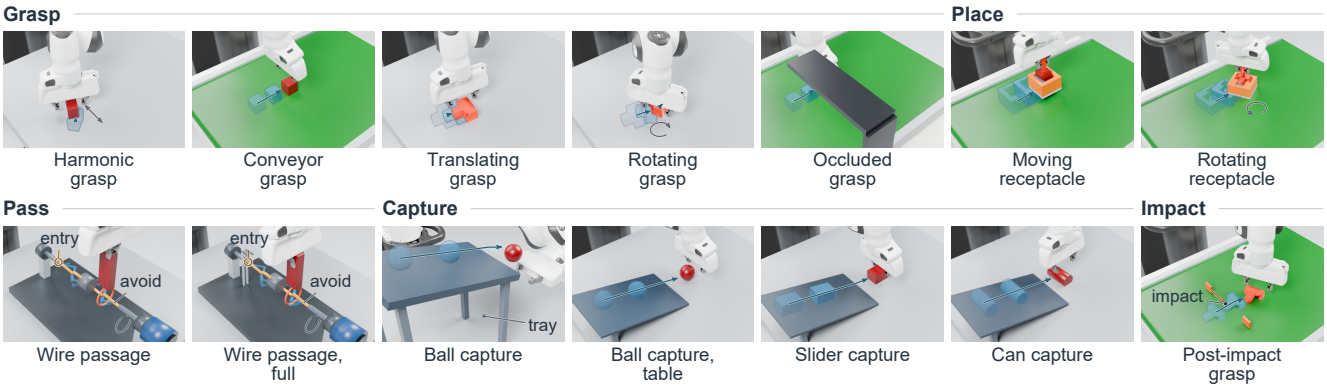


Figure 4. **OnTimeBench.** Fourteen tasks in five families, rendered from recorded simulator states with a common agent view. Translucent blue poses show selected history, fading with age; navy arrows follow recorded translational target motion. Local gray symbols denote rotation or oscillation qualitatively. An amber arrow shows the recorded striker approach; the *impact* label marks its contact location. Both Pass panels show passage: the wire is a target at entry and an obstacle along the path; Full additionally requires acquisition, exit and release. Occluded motion stays hidden. These native-expert examples illustrate task dynamics, not FoveaWAM outcomes; no future prediction is shown.

OpenWAM-α [39]; and the dynamic WAM DynamicWAM [25] (Table 1). Each baseline is trained on the same demonstrations with its official architecture and a task-adapted recipe. FoveaWAM receives a single agent-view RGB-D frame and a history of ground-truth SE(3) target poses from the simulator. To give baselines the same target-motion information, $\pi_{0.5}$ and DynamicWAM also run with our SE(3) history tokens (+SE(3)).

Implementation. The transformer of FoveaWAM has 12 layers of width 1024 with 16 heads and 271.8M trainable parameters on top of the frozen MAE ViT-B/16 encoder. On OnTimeBench, it trains from random initialization with fused AdamW for up to 20 epochs. We evaluate the final checkpoint of the seed-0 run after 10 epochs (25,870 updates). Evaluation uses compiled BF16 and five Euler steps with TF32 disabled. Compilation and recorded-observation warm-up finish before physics starts. We then reset the history and inference seed.

4.2. Dynamic Manipulation Results

Comparison with baselines. With every method charged its own latency, FoveaWAM reaches 82.1% SR on held-out OnTimeBench scenes (Table 1), 20.7 points above the strongest baseline, PUMA (61.4%; 95% paired interval [18.4, 23.1]). Target-motion input and low latency are both needed for this result. Given the same SE(3) history, $\pi_{0.5}$ and DynamicWAM gain 4.3 and 0.6 points and reach 58.7% and 41.7%; ACT, the only faster method, reaches 41.6%. FoveaWAM and ACT are the only methods whose queries finish within one 50 ms control period, and every other baseline hands over its chunk at least two periods after capture. FoveaWAM has the highest or tied-highest SR on 13 of the 14 tasks (Table S4). Its margins over the best baseline are largest where a moving receptacle or the target's orientation decides success: 30 and 55 points on placement into translating and rotating receptacles, and 28 points on full wire passage.

Table 1. **Results on OnTimeBench.** Task-balanced SR (%) on 1,400 held-out scenes with each method's native latency, with 150 ms added, and with physics paused during inference. p95: time-to-action in each method's native deployment setting on one RTX 5090. History: inputs that carry past motion beyond the current frame. OpenWAM-α uses its official caching and compilation, and (off) disables both. PUMA is timed with its official three-camera input; the other baselines with two cameras; FoveaWAM runs compiled BF16 inference on one RGB-D view and ground-truth target poses from the simulator. Best and second best per column are highlighted.

Method	History			SR (%) ↑			p95 (ms) ↓
	Wrist RGB	Flow	SE(3)	Native	+150 ms	Paused	
ACT [54]				41.6	24.6	40.6	**5.8**
DP [11]	✓			31.9	16.1	32.0	68.9
$\pi_{0.5}$ [5]				54.4	28.1	66.6	59.6
$\pi_{0.5}$+SE(3)			✓	58.7	28.0	76.0	61.2
DynamicVLA [42]	✓			51.1	23.5	54.9	97.3
PUMA [13]		✓		61.4	31.1	53.1	60.6
OpenWAM-α [39]				46.1	24.8	58.7	151.7
OpenWAM-α (off)				15.8	10.4	58.4	466.3
DynamicWAM [25]		✓		41.1	20.6	49.6	149.6
DynamicWAM+SE(3)		✓	✓	41.7	24.9	64.0	153.3
FoveaWAM			✓	**82.1**	**34.1**	**90.3**	15.1

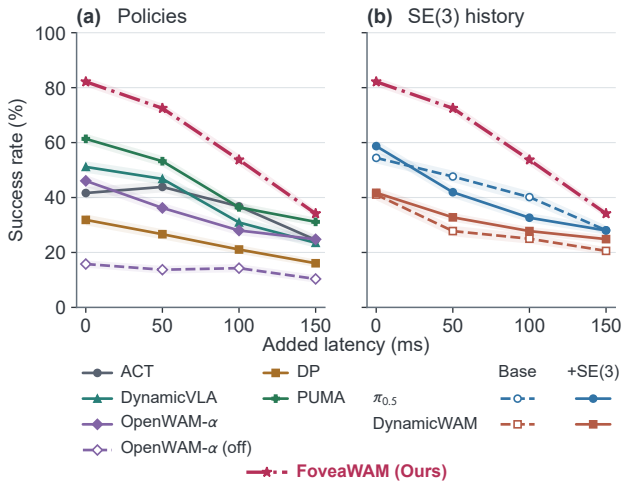


Figure 5. **Success under added latency.** Task-balanced SR on 1,400 held-out OnTimeBench scenes. Delay is added to each method's native latency, up to 150 ms, the largest step at which every method's chunk still covers the interval to its next chunk (Table S5 adds +200 ms). (a) Baseline deployments; off disables OpenWAM-α caching and compilation. (b) Base versus +SE(3)-history variants. PUMA is timed with its official three-camera input. FoveaWAM uses compiled BF16 (p95 15.1 ms). Bands: 95% task-stratified scene-bootstrap intervals.

Effect of inference delay. Latency alone can decide success. The two OpenWAM-α deployments run the same policy and agree with physics paused (58.7% and 58.4%), yet at native latency the one with a p95 of 151.7 ms reaches 46.1% and the one with 466.3 ms reaches 15.8%. Latency also erodes the value of motion input: the SE(3) history raises $\pi_{0.5}$ and DynamicWAM by 9.4 and 14.4 points with physics paused but by 4.3 and 0.6 at native latency. Charging latency therefore changes the ranking: $\pi_{0.5}$+SE(3), the strongest baseline with physics paused (76.0%), loses 17.3 points and falls behind PUMA. FoveaWAM leads in both settings, by 14.3 points over $\pi_{0.5}$+SE(3) and 20.7 over PUMA. Under added latency (Fig. 5), FoveaWAM remains the best method at every delay, ahead of the strongest baseline by 19.3 points at +50 ms, 13.6 at +100 ms, and 3.0 at +150 ms.

Results on DOMINO. To test the design beyond OnTimeBench, we train FoveaWAM, $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of DOMINO [13] with its demonstrations and evaluate them with its native evaluator. As on OnTimeBench, FoveaWAM reads ground-truth target poses from the simulator. This evaluator pauses physics during inference, so FoveaWAM gains nothing from its low latency here; it still reaches the highest overall SR, 32.5%, 5.6 points above the strongest baseline, DynamicWAM (26.9%), and the highest manipulation score (Table 2). It leads on the 16 Track tasks, 39.4% against 31.9% for DynamicWAM; PUMA is strongest on the 8 Intercept tasks at 36.3%. More results are in the supplementary material.

Table 2. **DOMINO results** on 24 single-arm Franka clean-L1 tasks, with the same 2,400 test scenes per method (100 per task). Intercept/Track contain 8/16 tasks; All weights all 24 tasks equally. SR and MS use a 0–100 scale and retain failures. Best and second best are highlighted.

Method	Intercept	Track	All ↑	MS ↑
$\pi_{0.5}$	23.8	8.1	13.3	27.8
DynamicWAM	16.8	31.9	26.9	41.4
PUMA	**36.3**	12.2	20.2	35.4
FoveaWAM	18.6	**39.4**	**32.5**	**49.1**

Table 3. **Analysis of role-aware modeling.** Task-balanced SR (%) of FoveaWAM on held-out OnTimeBench scenes and its change for each control (Δ, points). Red marks drops and green marks gains, with shading scaled within each column. Rot./Ctx.: rotation-sensitive and context-active tasks (Table S2). Table S7 adds paused physics.

	Success rate, all tasks		Groups (Native)	
	Native	+150 ms	Rot.	Ctx.
FoveaWAM	**82.1**	**34.1**	**80.4**	**82.7**
Precision allocation				
Uniform coarse flow	−4.5	−0.5	−8.6	−4.6
Uniform fine flow	+0.8	−0.9	−3.4	−2.6
w/o flow	−4.2	−1.4	−11.7	−3.9
Target motion history				
w/o history	−19.6	+0.2	−24.0	−2.3
Future supervision				
w/o future	−4.3	−3.4	−12.6	−2.1
w/o $\mathcal{L}_{\text{aux}}$	−2.6	−1.9	−11.6	+0.1

Table 4. **Real-robot results.** Success rate (%) on the six tasks of Sec. 4.4, 20 trials per task and method; All pools the trials. p95: end-to-end time-to-action (ms) on one RTX 5090, including tracking for FoveaWAM. Best per column is highlighted, including ties.

Method	T1	T2	T3	T4	T5	T6	All ↑	p95 (ms) ↓
$\pi_{0.5}$	—	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—	—
ACT	—	—	—	—	—	—	—	—
FoveaWAM	—	—	—	—	—	—	37.5	—

4.3. Analysis of Role-Aware Modeling

Six controls share the data, recipe, and executor of FoveaWAM (Table 3). The history control changes the input. The other five change only the supervision of prediction heads that are skipped at deployment, so they run the same

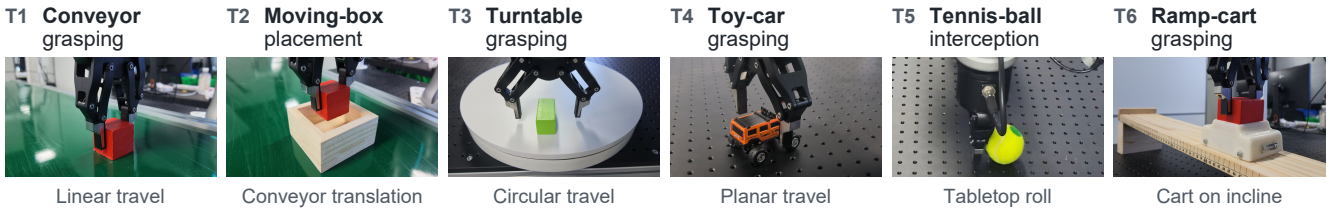


Figure 6. **Real-robot task setups.** Conveyor grasping (T1), placement into a moving box (T2), turntable grasping (T3), moving toy-car grasping (T4), rolling tennis-ball interception (T5, demo frame), and block grasping from a ramp cart (T6).

inference at the same latency as FoveaWAM, and any difference comes from what the model learned to forecast. We analyze SR at native latency, the condition FoveaWAM is designed for, overall and on the predefined rotation-sensitive and context-active task groups.

Allocation of precision. Role-aware supervision gives the target a precision that uniform supervision does not. At the same flow density, replacing the SE(3) trajectory with uniform flow reduces SR by 4.5 points overall and by 8.6 on rotation-sensitive tasks. Raising the uniform flow to a 56×56 grid, $16\times$ the points, costs [X.X]$\times$ the training time of FoveaWAM. It recovers the overall SR (+0.8) but still trails the role-aware model on both task groups, by 3.4 points on rotation-sensitive tasks and 2.6 on context-active tasks. One target trajectory and 196 flow points thus serve these tasks better than 3,136 uniform flow points, at a fraction of the training cost. The sparse flow is needed as well: removing it while keeping the SE(3) trajectory reduces SR by 4.2 points overall, by 3.9 on context-active tasks, and by 11.7 on rotation-sensitive tasks. The two forecasts are complementary, and FoveaWAM has the highest SR on rotation-sensitive tasks among all variants. All three controls keep the visual auxiliary objectives.

Target motion history. The target-motion history is the largest single factor. Keeping the current target pose but dropping its history, including the velocity and acceleration features, reduces SR by 19.6 points overall and by 24.0 on rotation-sensitive tasks. A single pose does not show how the target moves; the metric history supplies the motion that the commands must anticipate.

Future supervision and auxiliary objectives. Supervising the future, rather than only reconstructing the current target, is worth 4.3 points overall and 12.6 on rotation-sensitive tasks, and 3.4 points with 150 ms added, although the forecasting heads never run at deployment. The visual auxiliary objectives $\mathcal{L}_{\text{aux}}$ add 2.6 points overall and 11.6 on rotation-sensitive tasks. Every component of the role-aware design therefore contributes: removing any one of them lowers SR by 2.6 to 19.6 points overall and by 8.6 to 24.0 on rotation-sensitive tasks, where contact depends on the target's orientation.

4.4. Real-Robot Evaluation

Setup. The six tasks on a Flexiv Rizon 4s arm with a GN01 gripper include rolling tennis-ball interception (T5) (Fig. 6 and Table S8). Each method draws training data from a shared corpus of 600 teleoperated demonstrations. We plan 20 trials per task and method. FoveaWAM starts from a DROID-pretrained base [18]. In the planned evaluation, it uses RGB-D and online target pose estimates from the fixed camera only. The baselines ($\pi_{0.5}$, ACT, and DynamicWAM) use RGB from the fixed and wrist cameras.

Results. FoveaWAM succeeds in 37.5% of all trials, against [XX.X]% for [the strongest baseline] (Table 4). Its p95 latency, including tracking, is [XX] ms, [within one control period], whereas [name] needs [XX] ms. The margin is largest on [tasks]. [Whether the ranking of Table 1 carries over to real perception, and what tracking costs.] [One trajectory-verified contact example: how the target's pose, rotation, or context explains a success of FoveaWAM or a failure of a baseline.] The dominant failure of FoveaWAM is [failure mode]. Additional details are in the supplementary material.

5. Conclusion

FoveaWAM spends precision where contact depends on it: a metric SE(3) history and forecast for the target and a sparse 3D flow for the whole scene, with the forecasts used only as training supervision. When every method is charged its own latency, these choices let FoveaWAM act both precisely and quickly: it outperforms reactive policies, VLAs, and world–action models, including ones built for moving targets, on OnTimeBench and on six real-robot tasks at a p95 latency of 15.1 ms on OnTimeBench with compiled BF16 inference, and it also leads on DOMINO. The strongest baseline is PUMA under native latency and $\pi_{0.5}$+SE(3) in the paused column. At native latency on OnTimeBench, removing target pose history and removing future supervision reduce

success by 19.6 and 4.3 percentage points, respectively. We will release OnTimeBench so that future dynamic policies are evaluated with their latency charged.

Limitations. Deployment requires a specified rigid target and reliable online pose estimates. The target forecast predicts one future per horizon and ignores the effects of robot contact on target motion. Uncertainty-aware forecasts and contact-conditioned dynamics could address these limits. Independently moving context lies outside our task set and would need its own motion history, for example from scene flow.

LLM Usage Statement

We used OpenAI Codex to assist with literature review, code implementation and debugging, experiment execution, and language editing of the manuscript. Codex also assisted in developing plotting code and refining figure and manuscript layouts. The authors reviewed AI-assisted content, checked references against original sources, and validated the implementation through testing and experiments. The authors take full responsibility for the final manuscript, experimental results, and conclusions.

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